



```
In [5]: # Leap Year
import calendar

year = 2024

if calendar.isleap(year):
    print('Leap')

else:
    print('Not Leap')
```

Leap

```
In [8]: # 2nd Largest Number

n = int(input('Enter size of list : '))
l = []

for i in range(0,n):
    ele = int(input(f'Enter {i} index element : '))
    l.append(ele)

print('Before sorting : ', l)
l.sort()
print('After sorting : ', l)

print('\n2nd largest element : ', l[-2])
```

Before sorting : [2, 4, 1, 3, 9]
After sorting : [1, 2, 3, 4, 9]

2nd largest element : 4

```
In [2]: # Palindrome
word = input('Enter a word : ')

l = list(word)
print(l)

rev = l[:]
rev.reverse()
print(rev)

if(l == rev):
    print('Palindrome')

else:
    print('Not Palindrome')
```

['1', '2', '3']
['3', '2', '1']
Not Palindrome

```
In [28]: # Given Number is Prime or Not
n = int(input('Enter a number : '))
```

```

flag = True # Given Number is Prime

if(n==1):
    print('Neither Prime Nor Composite')

else:
    for i in range(2,n):
        if(n % i == 0):
            print('Not Prime')
            flag = False
            break

    if(flag):
        print('Prime')

```

Prime

```

In [30]: def isprime(n):
        if(n==1):
            return False

        else:
            for i in range(2,n):
                if(n % i == 0):
                    return False
            return True

```

```

In [36]: num = int(input('Enter a number : '))

        for i in range(2,num+1):
            if(isprime(i)):
                print(i)

```

2
3
5
7
11
13
17
19
23
29

```

In [50]: # List Duplicates
        l = [1,2,2,3,4] # 5
        # {1,2,3,4} - 4
        def isunique(l):
            s = set(l)

            if(len(l) == len(s)):
                return True

            else:

```

```

        return False

def numdup(l):
    s = set(l)
    return len(l)-len(s)

isunique(l)
numdup(l)

```

Out[50]: 1

```

In [48]: m = [5,6,7]
         isunique(m)

```

Out[48]: True

```

In [37]: # Membership Operator - in, not in
         l = [1,2,3]
         print(1 in l)
         print(2 in l)
         print(6 in l)

```

True
True
False

```

In [38]: # List Printing Methods
         m = [1,2,3,4]

         # 1st method - Index
         for i in range(0,4):
             print(m[i])

```

1
2
3
4

```

In [39]: # 2nd method
         for ele in m:
             print(ele)

```

1
2
3
4

```

In [46]: a = {1,2,3,3}
         a

```

Out[46]: {1, 2, 3}

```

In [54]: # Count each element in a list using dictionary
         a = [1,1,2,3,3,3,4,4]

```

```
def lcount(l):
    d = {}
    s = set(l)
    print(s)

    for ele in s:
        d[ele] = 0
    print(d)

    for key in d:
        for ele in a:
            if(ele == key):
                d[key] += 1
    print(d)

lcount(a)
```

{1, 2, 3, 4}

{1: 0, 2: 0, 3: 0, 4: 0}

{1: 2, 2: 1, 3: 3, 4: 2}

10

Write a program that takes a list of numbers and returns a new list with only the even numbers, squared.

X^2

```
In [55]: def iseven(num):
    if(num % 2 == 0):
        return True
    else:
        return False

    def sq(num):
        return num**2
```

```
In [64]: l = [2,3,4,6,5]

def newlist(l):
    res = []

    for ele in l:
        if(iseven(ele)):
            res.append(ele)

    for i in range(0,len(res)):
        res[i] = sq(res[i])

    print(res)

newlist(l)
```

[4, 16, 36]

In []: